

RT vs non-RT task threads - choreography-mqueue

Student and Welch t-tests on the per-iteration timings of perf.csv

Experiments compared

RT test_single_core_1 (fifo prio 50,60,70)

non-RT test_single_core_nort_2 (normal nice -8,-4,-1)

Method

99 RT and 99 non-RT iterations per task (first 1 iteration(s) dropped as warm-up); one test per task and metric.

Two-sided, unpaired tests; p-values Holm-corrected within each comparison; significance level $\alpha = 0.05$.

Welch's test is the reference (it does not assume equal variances); Student's is reported for comparison.

Key results: end-to-end response time, test_single_core_1 vs test_single_core_nort_2

Task 1: RT 88.1 ms vs non-RT 87.0 ms -> non-RT 1.04 ms faster (large effect), Welch $p < 0.001$

Task 2: RT 265.2 ms vs non-RT 184.1 ms -> non-RT 81.07 ms faster (large effect), Welch $p < 0.001$

Task 3: RT 88.5 ms vs non-RT 209.6 ms -> non-RT 121.17 ms slower (large effect), Welch $p < 0.001$

Task 4: RT 156.8 ms vs non-RT 190.9 ms -> non-RT 34.09 ms slower (large effect), Welch $p < 0.001$

Task 5: RT 177.5 ms vs non-RT 137.5 ms -> non-RT 39.99 ms faster (large effect), Welch $p < 0.001$

Task 6: RT 88.8 ms vs non-RT 125.6 ms -> non-RT 36.74 ms slower (large effect), Welch $p < 0.001$

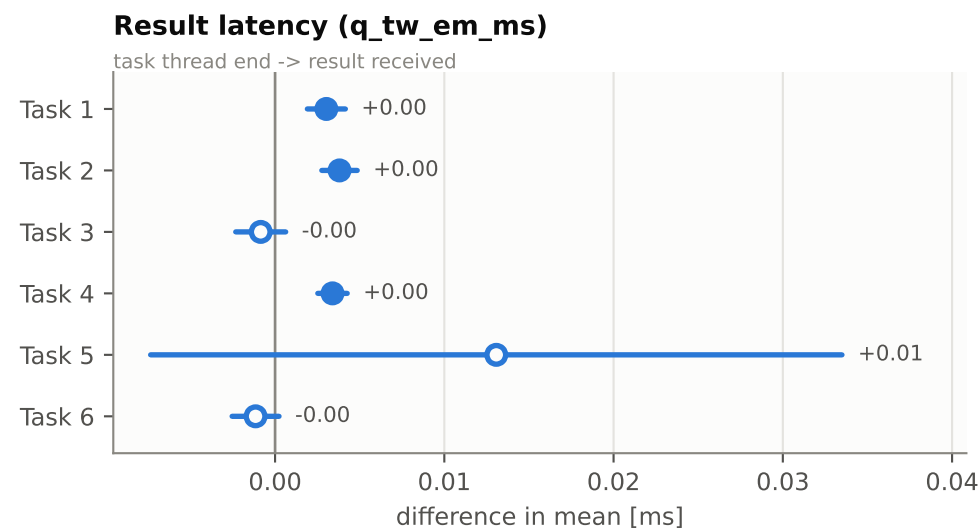
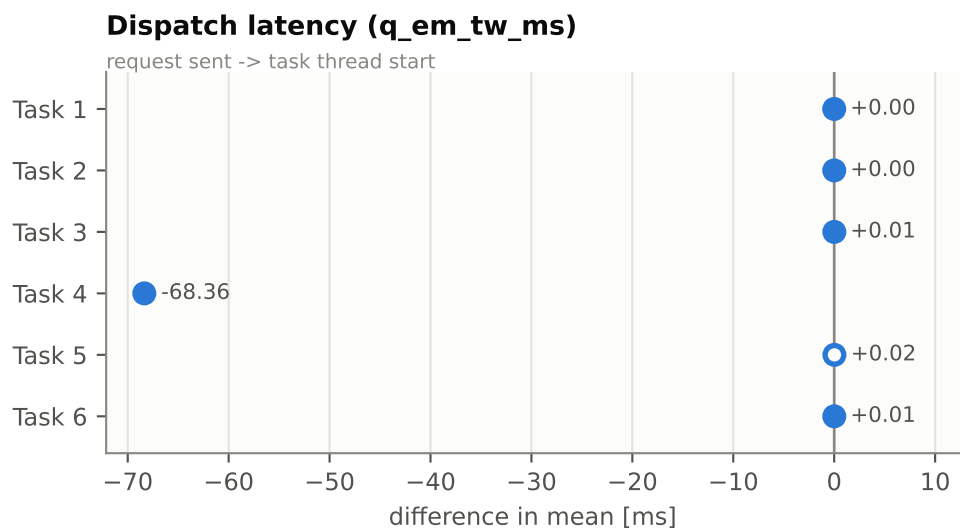
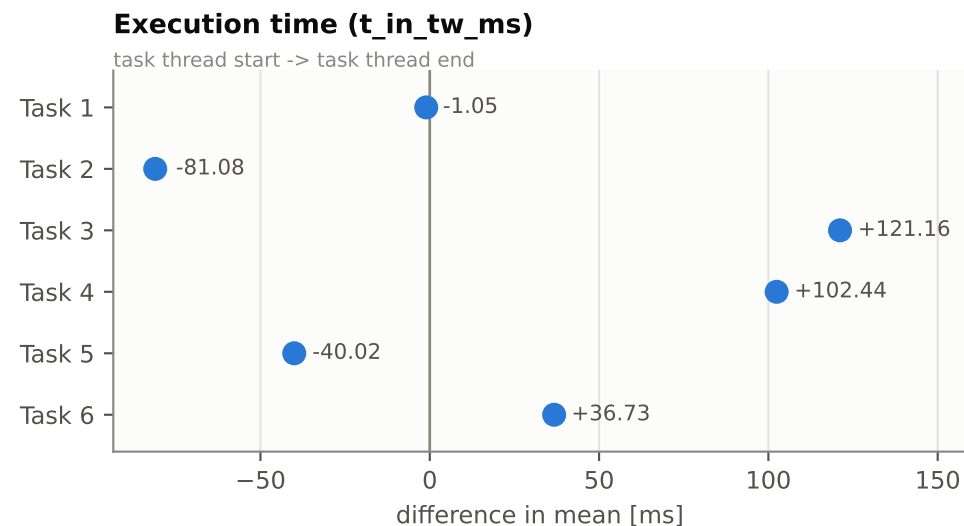
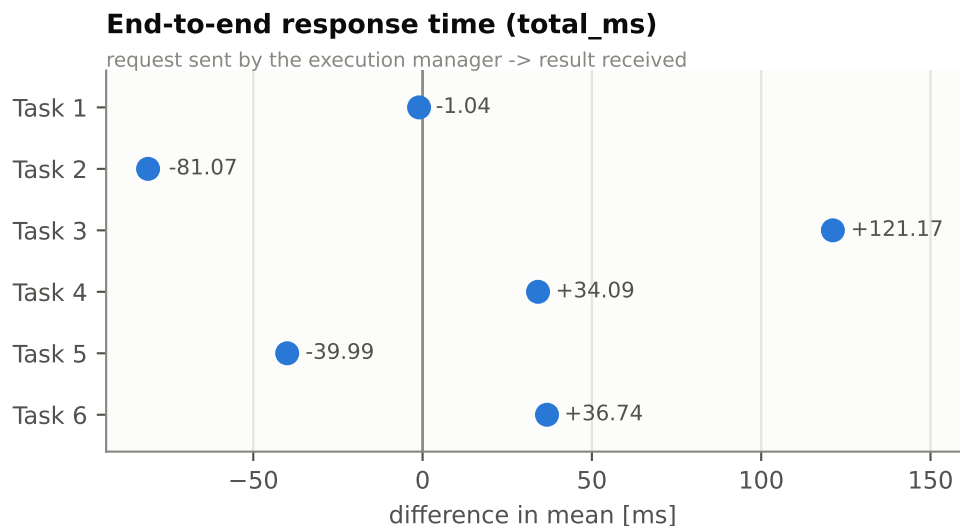
Overall: 20 of 24 task/metric tests significant (Welch); Student and Welch disagree on 0.

Note: with ~100 samples per side, even tiny differences are significant. Judge importance by the difference in ms and the effect size, not by the p-value.

Difference in mean time: non-RT minus RT

● significant (Welch, Holm) ○ not significant

test_single_core_1 (fifo prio 50,60,70) vs test_single_core_nort_2 (normal nice -8,-4,-1). Right of 0 = non-RT slower. Bars: Welch 95% confidence interval.

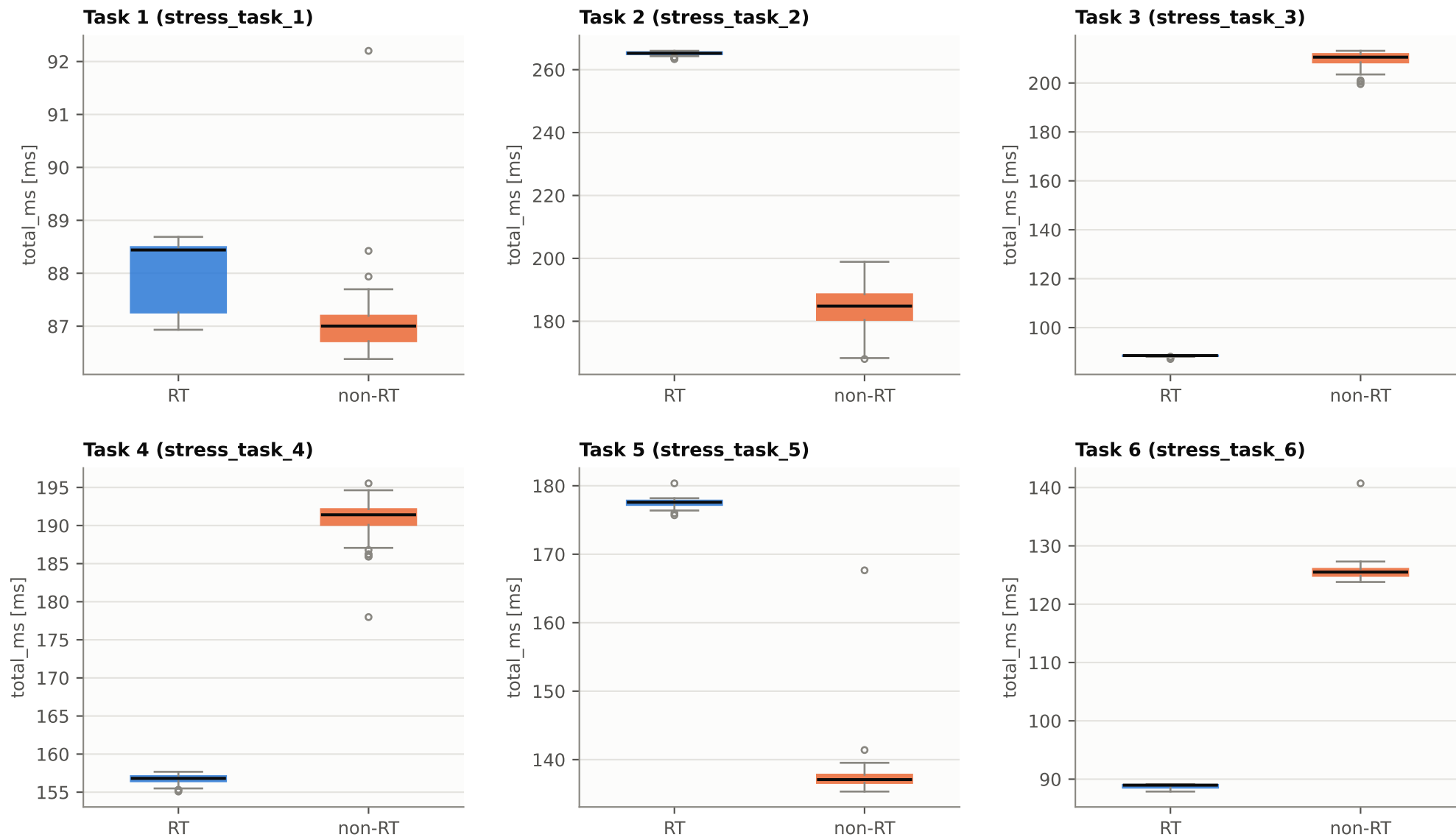


Distribution of end-to-end response time per task

RT (test_single_core_1) non-RT (test_single_core_nort_2)

test_single_core_1 (fifo prio 50,60,70) vs test_single_core_nort_2 (normal nice -8,-4,-1).

Box: 25th-75th percentile, line: median, whiskers: 1.5 IQR, dots: outliers. Each panel has its own scale.



All results

test_single_core_1 (fifo prio 50,60,70) vs test_single_core_nort_2 (normal nice -8,-4,-1). Means in ms ($\pm$ standard deviation); p-values Holm-corrected.

Task	Metric	RT mean $\pm$ sd	non-RT mean $\pm$ sd	Difference [95% CI]	Student p	Welch p	Hedges' g	Verdict (Welch)
1	total_ms	88.080 $\pm$ 0.625	87.039 $\pm$ 0.624	-1.042 [-1.217, -0.867]	< 0.001	< 0.001	-1.66 (large)	non-RT 1.04 ms faster (large effect)
1	t_in_tw_ms	87.984 $\pm$ 0.625	86.936 $\pm$ 0.621	-1.048 [-1.222, -0.873]	< 0.001	< 0.001	-1.68 (large)	non-RT 1.05 ms faster (large effect)
1	q_em_tw_ms	0.089 $\pm$ 0.005	0.092 $\pm$ 0.007	+0.003 [+0.001, +0.005]	0.002	0.002	0.52 (medium)	non-RT 0.00 ms slower (medium effect)
1	q_tw_em_ms	0.008 $\pm$ 0.004	0.011 $\pm$ 0.004	+0.003 [+0.002, +0.004]	< 0.001	< 0.001	0.75 (medium)	non-RT 0.00 ms slower (medium effect)
2	total_ms	265.153 $\pm$ 0.500	184.083 $\pm$ 6.793	-81.070 [-82.428, -79.712]	< 0.001	< 0.001	-16.77 (large)	non-RT 81.07 ms faster (large effect)
2	t_in_tw_ms	265.063 $\pm$ 0.500	183.984 $\pm$ 6.793	-81.078 [-82.437, -79.720]	< 0.001	< 0.001	-16.77 (large)	non-RT 81.08 ms faster (large effect)
2	q_em_tw_ms	0.084 $\pm$ 0.006	0.089 $\pm$ 0.008	+0.005 [+0.003, +0.007]	< 0.001	< 0.001	0.70 (medium)	non-RT 0.00 ms slower (medium effect)
2	q_tw_em_ms	0.007 $\pm$ 0.003	0.011 $\pm$ 0.004	+0.004 [+0.003, +0.005]	< 0.001	< 0.001	1.02 (large)	non-RT 0.00 ms slower (large effect)
3	total_ms	88.458 $\pm$ 0.234	209.629 $\pm$ 2.838	+121.171 [+120.603, +121.739]	< 0.001	< 0.001	59.96 (large)	non-RT 121.17 ms slower (large effect)
3	t_in_tw_ms	88.383 $\pm$ 0.233	209.541 $\pm$ 2.838	+121.158 [+120.590, +121.726]	< 0.001	< 0.001	59.94 (large)	non-RT 121.16 ms slower (large effect)
3	q_em_tw_ms	0.064 $\pm$ 0.005	0.078 $\pm$ 0.013	+0.014 [+0.011, +0.016]	< 0.001	< 0.001	1.37 (large)	non-RT 0.01 ms slower (large effect)
3	q_tw_em_ms	0.011 $\pm$ 0.006	0.010 $\pm$ 0.005	-0.001 [-0.002, +0.001]	0.612	0.616	-0.16 (negligible)	no evidence of a difference
4	total_ms	156.768 $\pm$ 0.515	190.857 $\pm$ 2.377	+34.089 [+33.605, +34.574]	< 0.001	< 0.001	19.74 (large)	non-RT 34.09 ms slower (large effect)
4	t_in_tw_ms	88.323 $\pm$ 0.267	190.765 $\pm$ 2.378	+102.442 [+101.965, +102.919]	< 0.001	< 0.001	60.32 (large)	non-RT 102.44 ms slower (large effect)
4	q_em_tw_ms	68.439 $\pm$ 0.418	0.084 $\pm$ 0.007	-68.356 [-68.439, -68.272]	< 0.001	< 0.001	-230.26 (large)	non-RT 68.36 ms faster (large effect)
4	q_tw_em_ms	0.005 $\pm$ 0.001	0.008 $\pm$ 0.004	+0.003 [+0.003, +0.004]	< 0.001	< 0.001	1.10 (large)	non-RT 0.00 ms slower (large effect)
5	total_ms	177.506 $\pm$ 0.543	137.520 $\pm$ 3.233	-39.986 [-40.639, -39.333]	< 0.001	< 0.001	-17.19 (large)	non-RT 39.99 ms faster (large effect)
5	t_in_tw_ms	177.417 $\pm$ 0.542	137.394 $\pm$ 3.052	-40.023 [-40.641, -39.405]	< 0.001	< 0.001	-18.19 (large)	non-RT 40.02 ms faster (large effect)
5	q_em_tw_ms	0.080 $\pm$ 0.005	0.104 $\pm$ 0.187	+0.024 [-0.013, +0.061]	0.612	0.616	0.18 (negligible)	no evidence of a difference
5	q_tw_em_ms	0.008 $\pm$ 0.005	0.022 $\pm$ 0.102	+0.013 [-0.007, +0.033]	0.612	0.616	0.18 (negligible)	no evidence of a difference
6	total_ms	88.843 $\pm$ 0.267	125.587 $\pm$ 1.733	+36.744 [+36.395, +37.094]	< 0.001	< 0.001	29.53 (large)	non-RT 36.74 ms slower (large effect)
6	t_in_tw_ms	88.762 $\pm$ 0.271	125.497 $\pm$ 1.729	+36.735 [+36.386, +37.084]	< 0.001	< 0.001	29.56 (large)	non-RT 36.73 ms slower (large effect)
6	q_em_tw_ms	0.071 $\pm$ 0.020	0.081 $\pm$ 0.022	+0.011 [+0.005, +0.017]	0.003	0.003	0.50 (small)	non-RT 0.01 ms slower (small effect)
6	q_tw_em_ms	0.010 $\pm$ 0.005	0.009 $\pm$ 0.004	-0.001 [-0.003, +0.000]	0.411	0.411	-0.23 (small)	no evidence of a difference

How to read this report

Metrics

total_ms End-to-end response time: request sent by the execution manager -> result received

t_in_tw_ms Execution time: task thread start -> task thread end

q_em_tw_ms Dispatch latency: request sent -> task thread start

q_tw_em_ms Result latency: task thread end -> result received

$\text{total_ms} = \text{q_em_tw_ms} + \text{t_in_tw_ms} + \text{q_tw_em_ms}$. Use total_ms for the overall effect: waiting can move between the parts.

Difference $\text{mean}(\text{non-RT}) - \text{mean}(\text{RT})$. Positive = non-RT slower, negative = non-RT faster.

95% CI range that contains the true difference with 95% confidence (Welch). If it does not cross 0, the difference is significant.

p-value probability of a difference at least this large if the true means were equal. < 0.05 = significant. ' < 0.001 ' means practically zero.

Holm correction for running many tests at once, so that 'significant' results are not found by chance.

Student vs Welch Student assumes equal variances, Welch does not (reference). A much smaller Student p signals that the variances differ.

Hedges' g difference in units of standard deviation: < 0.2 negligible, < 0.5 small, < 0.8 medium, otherwise large.

Caveats a non-significant result is not proof of equality; the t-test compares means only - check the distributions page for jitter and outliers.